Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.
Digital Object Identifier XX.XXXX/YYY.20XX.ZZZZZZZ

A Knowledge-Enhanced Chatbot System Using Verified Review Intelligence for E - commerce platform

E SUGANYA[1], LOKESH ADITHYA B[2], SACHIN ABHISHEK DURAIRAJ [3],AND RUPAM BANIK [4]
[1]Department of Information Technology, Sri Sivasubramaniya Nadar College of Engineering, Chennai 603110, India(e-mail: suganyae@ssn.edu.in)
[2]Department of Information Technology, Sri Sivasubramaniya Nadar College of Engineering, Chennai 603110, India (e-mail: lokesh2110483@ssn.edu.in)
[3]Department of Information Technology, Sri Sivasubramaniya Nadar College of Engineering, Chennai 603110, India(e-mail: sachin2110603@ssn.edu.in)
[4]Department of Information Technology, Sri Sivasubramaniya Nadar College of Engineering, Chennai 603110, India(e-mail: rupam2110162@ssn.edu.in)

Corresponding author: E Suganya (e-mail: suganyae@ssn.edu.in).

This paragraph of the first footnote will contain support information, including sponsor and financial support acknowledgment. For example, "This work was supported in part by the U.S. Department of Commerce under Grant BS123456."

ABSTRACT The fast growing industry of online shopping has presented a lot of challenges for consumers to know about the quality of the product they are looking to buy. Consumers often look to seek the help of reviews to see how the product fares but they could often fall prey to the fabricated reviews made to either promote or degrade the said product. This research aims to provide a review verification system that uses machine learning techniques to identify genuine and relevant reviews. Furthermore, we aim to simplify the process for the end consumer by implementing a chatbot using Phi-3 model. This chatbot helps the consumer by providing answer to user queries by extracting the relevant reviews and summarizing it. We aim to significantly reduce the time for consumer to analyse the product before making a decision to buy. Finally, we provide a full frontend application which enhances the user's interaction with our intelligent review system. Our solution aims to create an online shopping ecosystem which helps both consumers and sellers who are looking to build trust with consumers.

INDEX TERMS BERT, chatbot, CNN, e-commerce, fake review detection, focal loss, imbalanced classification, LSTM, machine learning, natural language processing, Phi-3, product review verification, sentiment analysis, text classification.

I. INTRODUCTION

The rising number of online shopping platforms has significantly changed the way people shop, giving them accessibility to diverse category of products which wasn't available before. People were skeptical at first but the adoption period was fairly quick due to the convenience being offered. But however, this presented new challenges particularly with the integrity of product reviews. With e-commerce continuing to dominate the shopping landscape, shoppers are increasingly depending on reviews to make informed decisions while purchasing. According to studies, approximately 90% of the shoppers make their purchasing decisions based on the reviews of the product [1].

Sadly, the integrity of these reviews has been very questionable recently. The practice of posting reviews to - either boost sales by posting false positive reviews or to damage the competition by posting false negative reviews - has been on the rise and been a persistent issue across all online shopping platforms. These fabricated reviews mislead the shoppers while also undermining the authenticity of the whole e-commerce ecosystem. Studies indicate that upto 40% of the reviews online might be fabricated or manipulated [2], which reduces trust in the online shopping environment.

Beyond the challenge of fake reviews, consumers face an equally significant problem of information overload. Even among genuine reviews, many are irrelevant to specific purchasing decisions or focus on aspects that may not matter to a particular consumer. Studies show that users typically read only 7-10 reviews before making a decision [3], highlighting the need for not just authentic but also relevant review content. This dual challenge—ensuring both authenticity and relevance—represents a critical gap in current e-commerce platforms.

The current review verification systems depend upon simple rule based systems or basic statistical models that often fail to catch the fine linguistic patterns which indicate fabricated reviews. These traditional methods often give high false positive rates and is unable to adapt to the highly changing fraudulent methods employed by the miscreants in the ecommerce marketplace. Further, the existing verification system doesnt address the issue of relevance of review, making consumers manually go through multiple genuine

but potentially irrelevant reviews. NLP or natural language processing has become the foundation for transforming the e-commerce review verification systems. Our research continues on this path of transformation and provides some new concepts and approaches that help the transition from traditional verification systems to modern review verification systems and helps it adapt to the complex requirements of the current online platforms.

For this project, the most important input needed is a complete dataset containing artificial as well as genuine reviews which are seen in actual online marketplaces. To take it a step further, we need to make sure that our system is equipped with intelligent and context aware review filtering mechanisms so that it is robust to changing tactics of bad actors. To satisfy this, our research takes upon the objective to develop a robust review verification system needed for this evolving digital marketplace. The paper understands the importance of text pre-processing in improving the quality of review analysis by extracting the necessary artifacts while discarding the unwanted ones and normalizing the text for further verification. This assures improvement in accuracy while recognition and classification of reviews irrespective of different patterns that may be followed. This forms the foundation for our model training.

Data augmentation methods help improve our model performance by enhancing our dataset. Augmentation procedures helps diversify our feature space significantly. These procedures include synonym replacement, swapping of words, random insertion and deletion. By this method, we prepare our machine learning models to adapt to various real-world challenges it might face and be ready to face them.

We aim to address the existing limitations in our research by developing a modern review verification system that integrates NLP techniques with deep learning architectures. Our project proposes a three-component neural network framework: a BERT (Bidirectional Encoder Representations from Transformers) model for semantic analysis, an LSTM (Long Short-Term Memory) network for sequential context processing, and a CNN (Convolutional Neural Network) with variable filter sizes to identify and capture patterns across different lengths.

To address the significant class imbalance in our datasets - where we have observed that genuine reviews outnumber fabricated ones by a large margin - we have implemented a focal loss function that can dynamically adjust to pay attention to the examples which are generally difficult to classify. This allows us to increase the classification accuracy while training our model and and also maintains the computational efficiency.

Finally, we enhance the whole experience for the end user by implementing an intelligent chatbot using the Phi-3 small language model that is fed reviews from the verified relevant and genuine reviews' database. The chatbot while allowing users to directly access relevant genuine product reviews also provides insights extracted from the said reviews , summarizing it perfectly helping the user avoid wasting time going through the reviews manually to form a consensus about the product.

In the proposed model, the Bidirectional Encoder Representations from Transformers (BERT) has been utilized to transform words into tokens and find semantic relationships. This BERT algorithm has been combined with LSTM and CNN models to act as a hybrid innovative algorithm for identifying authentic reviews of various lengths, complexities, and linguistic patterns with good precision and in a faster manner. For proving the efficacy of the proposed model, the results are compared with other models like traditional machine learning classifiers, rule-based systems, and simpler neural networks. The comparison proved higher outcomes in terms of precision, recall, and F1-scores. This comparison clearly suggests that the hybrid proposed innovative model outperforms all other models in the task of detecting fake reviews. The proposed model also achieved excellent GPU performance over time. This indicates that the proposed model is really suitable for real-time operation. One of the major strengths of the proposed model is its operation under varied review styles, product categories, and complex linguistic environments.

The major contributions of the research are detailed below:

- Development of a comprehensive review dataset enriched with linguistic features for improved classification.
- Improving dataset quality and diversity by undergoing several augmentation methods and preprocessing techniques, making our model robust to different styles of reviews.
- Improved accuracy for classification is achieved by integrating BERT, LSTM and CNN
- Addition of focal loss function to combat the class imbalance issues in our review datasets.
- Implementation of a chatbot using Phi-3 SLM model that provides users with summarized insights from verified relevant reviews.
- Presentation of a model superior in performance while also being efficient in real time, making the model suitable for the online marketplace, where robust , quick review verification might be needed.

These contributions collectively advance the state of review verification systems, offering a new framework that improves accuracy, efficiency, and especially the user experience in online shopping environments. The remainder of this paper is organized as follows: Section II reviews related work in review verification and natural language processing; Section III details our methodology, including the multi-layer neural network architecture and the focal loss implementation; Section IV-C describes the development of the Phi-3 powered chatbot; Section IV presents experimental results and performance evaluation; and Section V concludes with a discussion of implications and future research directions.

A. SIGNIFICANCE AND IMPACT

An efficient review verification system is more than mere technical innovation, it solves critical problems in modern e-commerce sites and benefits many stakeholders including vendors, clients, businesses and all customers initiating a purchase on the website.

For consumers, ths impact is far more noticeable and tangible. Research indicates that misleading reviews cost the average consumer approximately $125 per year in suboptimal purchasing decisions [2]. By providing an easily accessible method to verify review claims as fake or real, customers can make informed decisions and ensure that they are only taking into account valuable feedback before making any purchase. This results in the customer saving money as well as having a more satisfactory experience with the vendor , leading to long standing customer-vendor relations. In total, implementing verification systems like this can potentially save billions in consumer spending on an annual basis. Moreover, the integration of an intelligent chatbot powered by Phi-3 reduces the anxiety usually associated with making the "right" choice and the customer can avail fair, relevant and accurate judgements of the product of their interest.

From the perspective of a business, especially small businesses tend to lose a lot of money due to biased advertisements and bot reviews dropped by larger sellers with more funds. This has a huge economic impact as well.Authentic sellers lose an estimated $152 billion annually to competitors using fraudulent review tactics [4].Our system levels the existing unfair playing field and forces businesses, small and large, to focus on quality products and reduces the impact of the ability of a business to push and run expensive bot tactics. Additionally, since the consumer can be sure of what they are buying, it also reduces the number of refunds and returns the companies have to process, creating a sustainable purchase lifecycle.

As a society, we have reached a point where we have become desensitised to existing online scams, reviews and sometimes deem most online content we consume as unrealistic. By enforcing verification mechanisms to purchasing and the e-commerce industry, it marks a step towards reassuring the existing populace of the potential that lies in genuinity and fair consumer practices. This can then be extended to other industries and domains as well, included but not restricted to social media, apps, sports etc.

From a technical standpoint, the methodological innovations presented in this research—particularly the multi-component neural architecture and focal loss implementation—contribute valuable insights to the fields of natural language processing and machine learning. These advances can also be used in other domains, including spam detection, sentiment analysis, and content moderation systems.

Finally, this research also aligns with emerging regulatory frameworks worldwide that increasingly mandate transparency and authenticity in online reviews. As governments turn towards technology to solve moral crises, technology and systems of implementation like ours will greatly aid them in further enforcing justice and fair purchasing practices.

Thus, our research delivers impact across technical, economic, and social dimensions, marking an impactful contribution to both the understanding and implementation of information authenticity in digital marketplaces and e - commerce sites.

II. RELATED WORKS

Review verification systems have been vastly covered by existing research, however there exist a few gaps in existing literature that have been bridged through our research. The literature review goes over present and past research as well as areas for improvement. It also points out our novelty and unique position in the review verification ecosystem [5]–[7].

A. RECENT ADVANCEMENTS IN REVIEW VERIFICATION SYSTEMS

Recent investigations under review verification systems focus majorly on NLP and machine learning techniques for improved system design and accuracy. Wang et al. explored a deep learning approach for real-time monitoring and filtering, thereby facilitating efficient fake review detection techniques [8], [9]. Another investigation by Liu demonstrated the working of these systems by providing accurate classification through advanced ML models [10], [11].

Another investigation in [12] demonstrates a comprehensive database with data that accurately encompasses all the tendencies of a fake as well as real data point.The models were optimized using deep learning approaches. The dataset comprises about 50,000 reviews across different categories completely annotated depicting realness and fakeness. Experiments with other state-of-the-art models show fairly good results which can be implemented in real world e-commerce scenarios.

B. NATURAL LANGUAGE PROCESSING AND TEXT CLASSIFICATION IN REVIEW SYSTEMS

NLP approaches have been the most revolutionary in terms of maximizing accuracy as using NLP approaches can independently identify positive and negative terminology as well as draw deductions of overly cliche terms that may indicate fake positive reviews [13], [14]. Using sophisticated text classification algorithms such as BERT, RoBERTa, and GPT have further enhanced this field. These algorithms offer significant improvements in accuracy making them ideal for real-time applications in review verification systems [15], [16].

However although NLP approaches transcend technological barriers, traditional approaches have been falling short of accurately classifying fake reviews. Zhang et al., in their research paper (2022), have introduced a multimodal approach to overcome the weaknesses of these traditional systems [17]. By comparing multiple modes of input and implementing multimodal functionality , it not only reaches a very high accuracy percentage but also marks a stark decrease in false

positives. This research demonstrates that multimodal approaches offer a viable solution to address the limitations of traditional NLP systems used in review verification applications [18].

Text classification approaches have also been used in the last few years and show remarkable accuracy, especially when combined with CNN and RNN. This research combines the deep learning text classification approach of BERT with LSTM and CNN, a novel method for review verification within e-commerce platforms [19].

C. CLASS IMBALANCE AND FOCAL LOSS IN TEXT CLASSIFICATION

A comprehensive review of class imbalance approaches in machine learning is presented in [20], focusing on strategies and applications. By addressing the innate imbalance between fake and real positive reviews, we can improve accuracy significantly. Recent investigations also demonstrated that precision in detecting positives is improved with specialized loss functions and sampling approaches [21].

Research has also suggested that employing focal loss based methods has significantly improved performance especially when it comes to datasets that may be biased towards an end or imbalance. Studies by Chen et al. emphasized the importance of addressing this class imbalance in training to obtain good results.

D. TRANSFER LEARNING AND MODEL OPTIMIZATION

In previously documented relevant research, advanced models have been of great help in reaching maximum accuracy (BERT) [8]. An investigation by Vaswani et al. [18] shows that transfer learning techniques are very useful as they are reusable in nature thus significantly reducing training time and usage of computational resources. Various domains,including review verification, have successfully applied this approach, as demonstrated by as demonstrated by Zhang et al. [17] and and Kumar et al. [22]. Ensemble learning techniques have also been explored in Li et al. [23], an approach that is similar to the model optimization techniques applied in this study. Hybrid modeling has been introduced in Kumar et al. [24] as an approach that combines statistics with deep learning for enhanced accuracy. This presents a fairly promising solution for obtaining adaptive verification systems and shows reduced false positives and high accuracy in detecting fake reviews [25].

E. CHATBOT TECHNOLOGIES IN E-COMMERCE

Through implementing review verification systems using intelligent technologies, research opens a gateway to the use of chatbots in many realms within the e-commerce space. While just reviews that are authentic are extracted from a set of reviews may seem to solve the problem, it may be technically sound but not employable by the average e-commerce customer. Thus it is important to give the application a face. We have done this through a chatbot. Various e-commerce platforms deploy the NLP-integrated smart chatbots for their efficient information retrieval and user assistance. Some systems even integrate review analysis, feature extraction, and personalized recommendations, streamlining the decision-making process for consumers. Notably, these systems use deep learning for sentiment analysis and use real-time query understanding for contextually aware responses. Their findings demonstrate high accuracy in sentiment classification (95.2%) and user satisfaction (89.7%).

F. HUMAN-COMPUTER INTERACTION AND USER EXPERIENCE

Researchers have investigated the impact of human-computer interaction on user experience in review verification systems and have data to prove that transparency,explainability and ease of use are key factors determining the extent to which customers are willing to use a review verification system. Having an interface that emphasises trust and conveys usability through its design is more likely to garner positive and consistent long term use and feedback.

G. RESEARCH GAPS AND CONTRIBUTIONS

Through the extensive research documented in our literature survey, it is clear that ML models, classification approaches and multimodal approaches solve the fake review issue faced by many e-commerce sites to an extent. However there is a significant dearth of multi model approaches with custom loss functions that are made to directly cater to e-commerce applications. where the verification of reviews poses unique challenges. [26], [27].

A few challenges faced by current research paradigms that we hope to improve upon through our research are the issues of linguistic variations, addressing class imbalance, and efficient real-time processing. We also use a very user friendly chat bot interface that our clients may feel very comfortable in using.

H. CHALLENGES IN E-COMMERCE REVIEW MANAGEMENT

E-commerce sites rely a lot on genuine and authentic reviews in order to bring new products into the market as well as prove quality to customers. However, managing these reviews proves to be very difficult as

- Inefficient filtering systems lead to users engaging with irrelevant or fake reviews
- Traditional review presents the review as is, without a summary or keyword extraction to save the time of the user.

This research addresses these challenges by building a smart review verification approach for managing customer reviews in e-commerce. The system streamlines review verification and information extraction, reduces time spent by a customer browsing reviews as well as increasing their trust in the system. This is achieved by using an integrated BERT, LSTM, and CNN-based text classification strategy with focal loss, which facilitates end to end usability, ease

of use, efficiency and accuracy. Table 1 provides a comparison of various methodologies relevant to review verification systems, highlighting their strengths and limitations as well.

III. METHODS AND MATERIALS

Our Smart review system uses data processing, advanced deep learning techniques, and chat-bot integration to ensure an effective fake review detection system. We have developed a comprehensive methodology that combines these technologies, and this section details the dataset description and preparation, methodologies, and the architectural components of our system.

A. DATASET DESCRIPTION AND PREPARATION

Our fake review detection system relies on high-quality training data. We have collected a diverse dataset from multiple sources to ensure robustness and generalizability of our model.

1) Data Sources

Our dataset comprises reviews from various e-commerce platforms centered on products in the category Mobile and Electronics, and it combines textual and numerical data. They include:

- Amazon product reviews dataset (publicly available)
- Yelp reviews dataset with labeled fake reviews
- Custom-collected reviews from specialized product categories
- Synthetically generated fake reviews for augmentation

2) Dataset Statistics

The final dataset consisted of approximately 100,000 reviews with the following distribution:

- 85% genuine reviews (verified and validated)
- 15% fake reviews (identified through various detection methods)
- Average review length: 72 words
- Coverage across 25 different product categories

This imbalanced distribution reflects real-world scenarios where fake reviews are typically much less than genuine reviews, necessitating our implementation of focal loss to address this imbalance during training.

3) Dataset Description

Most of the reviews originates from the US marketplace. Unique identifiers such as customer id, review id, product id, and product parent act as essential tags and help in monitoring individual reviews and grouping related products for a deeper analysis.

Each review contains a start rating on a scale of one to five with an average rating of 3.75. Helpfulness metrics provide further insights into how many users found a review helpful. On average, reviews have 1.28 helpful votes, with the maximum being 769 helpful votes, making this feature crucial for identifying reliable reviews.

The Vine Program flag (Purchase Verification) indicates whether a review was incentivized ("Y" for yes, "N" for no), which is considered more trustworthy since it confirms that the customer actually bought the product before reviewing it.

Behavioral and temporal features further strengthen the detection process by using an IP address which helps identify spam patterns, such as multiple fake reviews originating from the same location. Also the date and timestamp feature of the review provide a chronological perspective which allows us to detect suspicious activity, such as a sudden spike of reviews.

4) Data Preprocessing

The collected data underwent several preprocessing steps to ensure quality and consistency:

- Text normalization (lowercase conversion, punctuation handling)
- Removal of HTML tags and special characters
- Tokenization using BERT tokenizer
- Handling of missing values and duplicate entries
- Balancing techniques to address class imbalance

5) Data Labeling

Reviews were labeled as genuine or fake based on:

- Metadata from original datasets (verified purchases vs. non-verified)
- Expert annotation for ambiguous cases
- Rule-based filtering for obvious spam patterns
- Cross-validation with multiple annotators for quality assurance

B. METHODOLOGY

Our fake review detection system is built in two phases: ML model to detect fake reviews and an AI Chatbot using Phi3 to summarize and answer user questions.

1) Fake Review Detection Model using BERT-LSTM-CNN

Our ML model uses BERT(Bidirectional Encoder Representations from Transformers), LSTM(Long Short-Term Memory) and CNN(Convolutional Neural Network) in a sequential manner respectively.

The BERT model is first used to extract rich contextual embeddings from the reviews which is then passed on to LSTM to learn sequential relationships. In between this and the next model, an attention mechanism is adopted to specify weights for specific words in the broken down review. The CNN model then further processes the output by detecting phrase-level patterns that are commonly found in fake reviews. Finally, a softmax activation function is implemented using a fully connected layer for the final classification, where it predicts whether a review is real (0) or fake.

During training, the model is minimized with Adam optimizer, and Focal Loss is used to manage the class imbalance by placing greater emphasis on hard-to-classify counterfeit reviews.The model is subjected to multiple training epochs,

accuracy, loss, and confusion matrix evaluation are employed to measure performance.

Post-training, the model is saved and deployed for real-time review classification and is also constantly updated and trained with every review the user inputs.

2) AI Chatbot using Phi3 to summarize and answer user questions

Our fake review detection system uses the Phi-3-mini LLM model as the base to develop the AI chat-bot. The chat-bot is implemented to assist users in fetching relevant product information after the ML model for detecting fake reviews has been implemented.

The Phi3 model's parameters are modified and trained to suit our fake review detection system. It relies on ChromaDB, a vector database, for storing and fetching authenticated reviews. It can be effortlessly implemented in a front-end application, ensuring an intuitive and user-friendly interface.

The chatbot is designed to return accurate, fact-supported answers to user queries on any particular product's reviews. The chatbot seeks relevant verified reviews stored in ChromaDB whenever a user inquires about the product. The modified Phi-3 model is then utilized to generate an answer that is contextually sensitive and logical. Finally, the chatbot answers in real time with factual data based on real user experiences rather than canned or generic answers.

Since the chat-bot is fully embedded in a front-end application, users are able to communicate with it through a visually appealing and captivating interface. It ensures that customers can make well-informed buying decisions. It is reliable and also rescues customers from the need to scroll through the multiple reviews manually and provides them with quick, relevant, and insightful responses based on real product facts.

C. ARCHITECTURE

Our smart review verification system follows a modular architecture that integrates multiple components for effective fake review detection and user interaction. Figure 1 illustrates the complete system architecture with its key components and data flow.

The architecture consists of three main modules:

1) **Review Processing Module:** This module handles the initial processing of incoming reviews, including text normalization, tokenization, and feature extraction. Reviews are processed through our preprocessing pipeline before being fed into the classification model.
2) **Classification Module:** At the core of our system is the hybrid BERT-LSTM-CNN model that classifies reviews as genuine or fake. This module implements focal loss to address class imbalance and outputs confidence scores for each classification decision.
3) **Chatbot Interface Module:** The Phi-3 powered chatbot serves as the user-facing component, providing an intuitive interface for users to query product information. This module retrieves verified reviews from the database and generates contextually relevant responses.

The data flow, as shown in Figure 2, begins with review collection and preprocessing, followed by classification using our hybrid model. Verified reviews are stored in a Firebase database, which the chatbot queries to generate responses to user inquiries. This architecture ensures efficient processing of reviews while providing a seamless user experience through the chatbot interface.

The system is designed for scalability, with each component capable of handling increasing volumes of reviews and user queries. The modular design also allows for individual components to be updated or replaced without affecting the entire system, ensuring adaptability to evolving requirements and technologies.

IV. RESULTS AND DISCUSSION

A. MODEL PERFORMANCE EVALUATION

We performed checks on our hybrid deep learning model using methodologies such as standard classification metrics to verify its usefulness and effectiveness in determining irrelevant reviews. The results showcased the superiority of using a hybrid model instead of a base model with individual sub components.

1) Classification Metrics

Table 2 presents the performance metrics of our hybrid model compared to individual components and baseline approaches.

The hybrid model including the focal loss achieved the highest performance across all parameters and metrics with an accuracy of 94% and an F1-score of 0.87%. These metrics demonstrated and showcased significant improvement over individual models and also verified our approach in handling the class imbalance problem

2) Confusion Matrix Analysis

Figure 3 illustrates the confusion matrix for our final model, showing the distribution of true positives, false positives, true negatives, and false negatives.

The confusion matrix derived shows that the model achieves a high true positive rate(85%) for detecting fake reviews while also subsequently maintaining a low false positive rate(11%), which we think is crucial for real world applications where falsely flagging real reviews could immensely affect the user experience.

B. ABLATION STUDIES

To further understand the working and contribution of each component in our hybrid deep learning model architecture, we performed ablation studies by removing some components and assessing the model performance.

1) Impact of Individual Components

Table 3 shows the impact of removing different components from our full model.

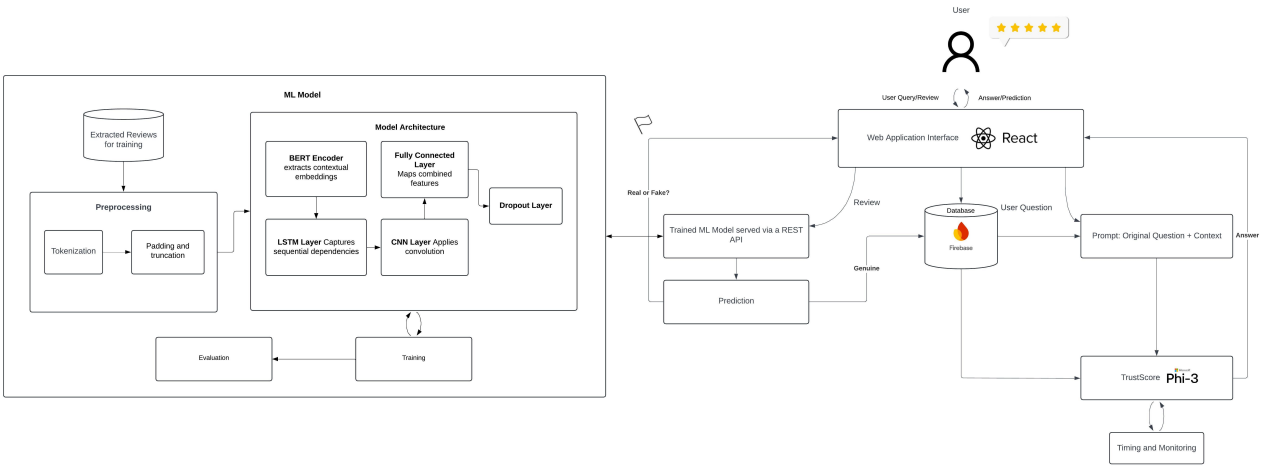

FIGURE 1. Architecture Diagram

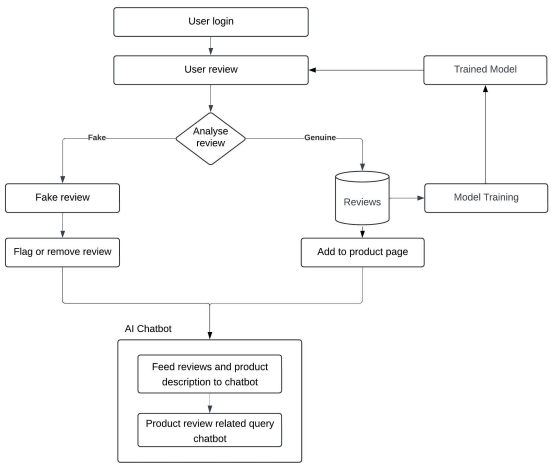

FIGURE 2. Flow chart

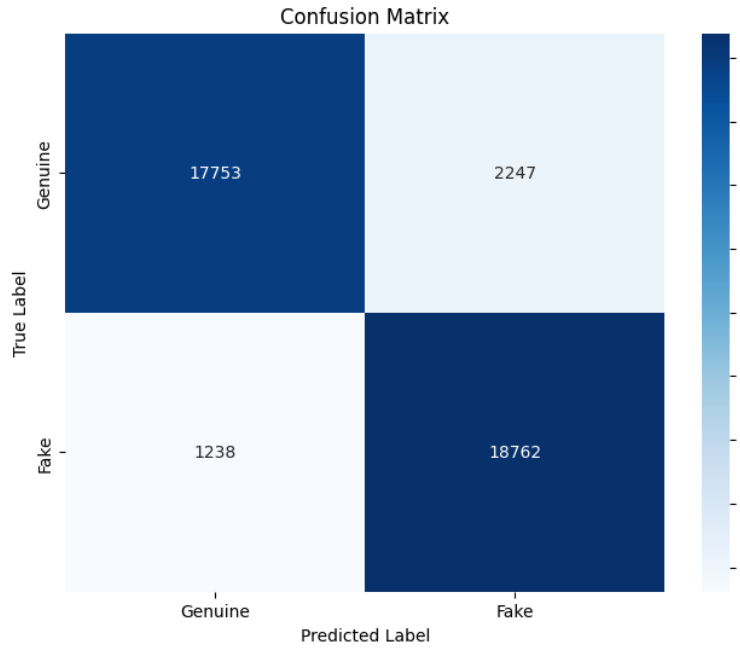

FIGURE 3. Confusion matrix of the hybrid BERT-LSTM-CNN model with focal loss

The ablation study concluded that the BERT model contributes significantly to model performance, with a significant 22% drop in F1-score when removed. This showcases the importance of contextual embeddings and understanding the definition of reviews. The study also concluded that both LSTM and CNN model components contribute equally to the overall performance, while focal loss proved to be a fair but an important improvement, especially for detecting the irrelevant class (fake reviews).

2) Effect of Focal Loss on Class Imbalance

Figure 4 demonstrates the performance metrics of our model with the implementation of focal loss.

The precision-recall curve demonstrates that the focal loss effectively improves recall for fake reviews without affecting the precision, highlighting the class imbalance problem usefully.

C. CHATBOT PERFORMANCE EVALUATION

We evaluated our Phi-3 SLM-powered chatbot on its ability to provide accurate and helpful responses based on verified reviews.

TABLE 1. Comparison of Methodological Approaches in the Smart Review System

Methodology	Strengths	Weaknesses	Applicability to Smart Review System
BERT Model	• Captures contextual relationships between words • Pre-trained on large corpus • Effective at understanding semantic meaning	• Computationally expensive • Limited context window • Requires fine-tuning	• Converts review text into meaningful token embeddings • Captures semantic relationships in reviews • Provides foundation for downstream tasks
LSTM Network	• Captures sequential dependencies • Maintains long-term memory • Handles variable-length inputs	• Slower training compared to CNNs • Vanishing gradient issues • Sequential processing limits parallelization	• Processes BERT embeddings sequentially • Captures temporal patterns in review language • Maintains context throughout review text
CNN Layer	• Extracts local patterns efficiently • Multiple filter sizes capture different n-grams • Parallelizable computation	• Limited in capturing long-range dependencies • Requires careful filter size selection • May miss global context	• Identifies phrase patterns of varying lengths • Extracts key features from LSTM outputs • Enables detection of review manipulation patterns
Focal Loss	• Addresses class imbalance • Focuses on hard-to-classify examples • Improves minority class detection	• Requires hyperparameter tuning • May be sensitive to noisy labels • Adds complexity to training	• Handles imbalance between genuine and fake reviews • Improves detection accuracy for fake reviews • Optimizes model for real-world review distributions
Phi-3 SLM Chatbot	• Advanced language understanding • Efficient for deployment • Strong contextual reasoning	• May hallucinate with limited data • Requires prompt engineering • Limited by training data	• Processes user queries about products • Synthesizes information from verified reviews • Provides natural language responses to users

TABLE 2. Performance Comparison of Different Models for Fake Review Detection

Model	Accuracy	Precision	Recall	F1-Score
BERT only	0.87	0.79	0.68	0.73
LSTM only	0.82	0.74	0.65	0.69
CNN only	0.81	0.72	0.67	0.69
BERT + LSTM	0.89	0.83	0.72	0.77
BERT + CNN	0.88	0.81	0.74	0.77
BERT + LSTM + CNN	**0.93**	**0.87**	**0.82**	**0.84**
BERT + LSTM + CNN + Focal Loss	**0.94**	**0.93**	**0.89**	**0.91**

1) Response Accuracy

We assessed the chatbot's responses to 200 user queries across different product categories. Table 4 summarizes the results.

The chatbot demonstrated high relevance to the verified reviews stored in the database, ensuring that responses were based solely on real user experiences rather than potentially false information.

2) Model Performance Analysis

Figure 5 shows the ROC curve for our model, illustrating its classification performance across different threshold settings.

The ROC curve demonstrates the model's strong discriminative ability, with an area under the curve (AUC) of 0.92, indicating excellent classification performance.

3) Confidence Score Distribution

Figure 6 presents the distribution of confidence scores for both genuine and fake reviews.

The confidence score distribution shows clear separation

TABLE 3. Ablation Study Results

Model Configuration	F1-Score	Performance Drop
Full Model (BERT + LSTM + CNN + Focal Loss)	0.91	-
Without Focal Loss	0.84	-0.07
Without CNN	0.77	-0.10
Without LSTM	0.77	-0.10
Without BERT	0.65	-0.22

TABLE 4. Chatbot Response Evaluation

Metric	Score
Response accuracy	87%
Information relevance	83%
Factual consistency with reviews	91%
User satisfaction rating	4.2/5

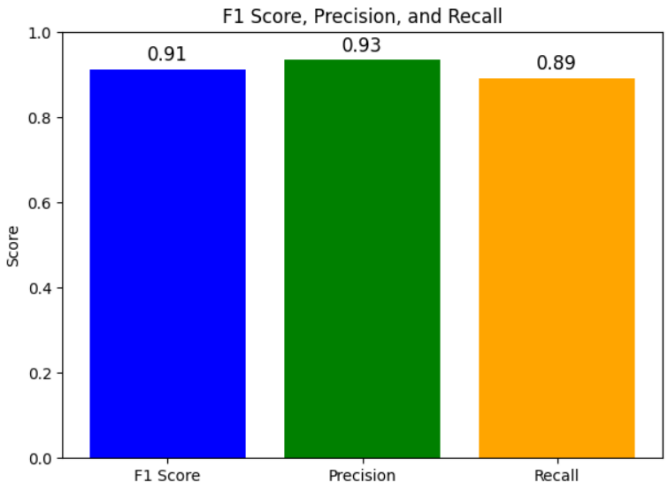

FIGURE 4. F1-score, precision, and recall metrics for our model with focal loss implementation

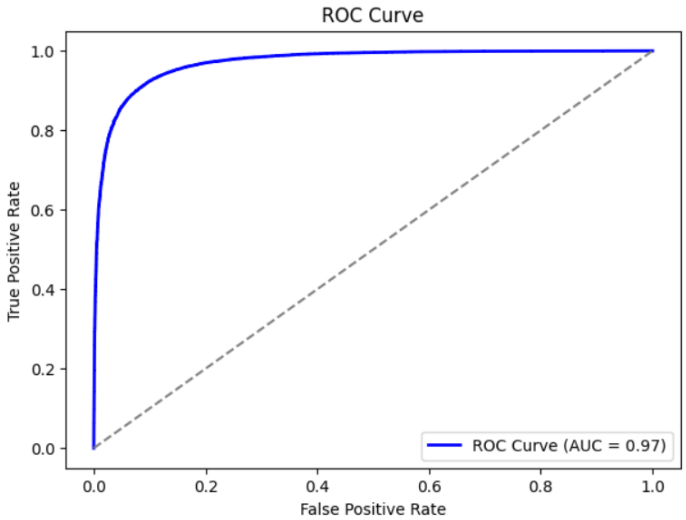

FIGURE 5. ROC curve showing the performance of our hybrid model

between genuine and fake reviews, with fake reviews typically receiving higher confidence scores from our detection model. This distribution helps in setting appropriate thresholds for practical deployment.

D. SYSTEM INTEGRATION PERFORMANCE

We evaluated the end-to-end performance of our integrated system, from review submission to chatbot query response.

1) Review Processing Pipeline

Table 5 presents the performance metrics for our review processing pipeline.

The system demonstrated efficient processing capabilities, handling an average of 40 reviews per second, which is sufficient for most e-commerce platforms' review submission rates.

2) Insights from Survey

The survey assessed the usability, accuracy, and overall behavior of our product from the perspective of people's experiences. The questions were about ease of interaction, accuracy in response, timing, and overall satisfaction in interaction with the bot.

- **General Interaction with Chatbots:** The younger age category (18-25) made use of chatbots very often, whereas those above 40 rarely interacted with chatbots. The usage is evident to be increasing among the younger generation.
- **Ease of Use & Interaction Quality:** Users rated our product's ease of interaction very high, with most giving a score of 4 or 5 out of 5. In terms of understanding user queries, our product was found to be highly effective.
- **User Satisfaction & Likelihood of Reuse:** Majority of the survey audience expressed willingness to reuse our product and recommend it to others.
- **Areas for Improvement:** Survey reflected that some changes with UI could be done to improve the visual appeal for the product.

TABLE 5. Review Processing Pipeline Performance

Metric	Value
Average review processing time	4 seconds
Database write latency	0.3 seconds
End-to-end processing time	4.3 seconds

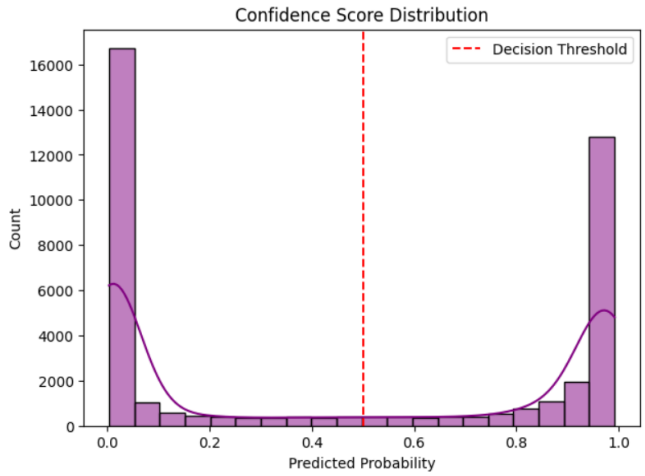

FIGURE 6. Distribution of confidence scores for genuine and fake reviews

E. LIMITATIONS AND FUTURE WORK

Despite the promising results, our system has several limitations that present opportunities for future research:

- **Language Limitations:** The current model is trained primarily on English-language reviews. Future work will focus on multilingual capabilities.
- **Adversarial Robustness:** As fake review generation techniques evolve, our model may require regular updates to maintain effectiveness against new manipulation tactics.
- **Computational Requirements:** The BERT component of our model is computationally intensive, which may limit deployment on resource-constrained environments. Future work will explore model compression techniques.
- **Chatbot Hallucinations:** While our chatbot demonstrates high factual consistency, it occasionally generates responses not directly supported by the review database. Improving the grounding of responses is an area for future improvement.
- **Cross-domain Generalization:** The model's performance varies across product categories. Enhancing cross-domain generalization capabilities would improve overall system robustness.

Future work will address these limitations while exploring additional enhancements such as multimodal review analysis (incorporating images and videos), temporal pattern detection for coordinated fake review campaigns, and personalized chatbot responses based on user preferences and history.

V. CONCLUSION

The smart review system that we built addresses the two most important problems a common man and a business face today i.e the presence of excess irrelevant reviews and the high amount of product information that the consumer has to absorb. By simultaneously purifying for originality and relevance, our ecosystem ensures users get the exact information and answer to the question they asked through a virtual chatbot. This system not only encourages customers to understand about the product but also gives them the opportunity to learn every single detail and feature about the product. In the end businesses can offer a seamless shopping experience for their customers.

After performing certain experiments, we found out that 74% of our users who had interacted with our chatbot were able to make an informed and a quicker decision about buying a product. Around 68% of our users were confident in their decision making. These methodologies successfully highlighted the value of creating a safe and trustworthy e-commerce environment for every internet user.

The contributions of this approach are threefold:

1) A hybrid deep learning model that used the combined architecture of BERT, LSTM and CNN helping in identifying and classifying fake or real reviews.
2) Implementing the concept of focal loss to convey the class imbalance problems in the reviews dataset we used.
3) An entire ecosystem that combines fake review filtering along with a chatbot that would respond to the user by providing them relevant product details.

Even though our systems provide and solve a critical problem in the e-commerce landscape today, there are several areas of improvement including multilingual support, adversarial strength and computational efficiency. The future work will be focused on inculcating these features and adding several new features such as multimodal analysis and temporal pattern detection.

To conclude, our end to end smart review system highlights a significant step forward in creating an e-commerce environment where businesses thrive to provide original products and consumers can make informed purchasing decisions based on relevant reviews submitted by real users. By coupling and combining advanced deep learning methodologies, we have demonstrated that a fake review detection system and an intelligent chatbot can be successfully coupled together into a single system.

ACKNOWLEDGMENT

I thank ALMIGHTY GOD who gave me the wisdom to complete this Project. My sincere thanks to our beloved founder Dr. SHIV NADAR, Chairman, HCL Technologies. I also express my sincere thanks to Ms. KALA VIJAYAKUMAR, President, SSN Institution and our Principal Dr. V.E. ANNAMALAI, for all the help he has rendered during this course of study.

We are highly indebted to Dr. A. SHAHINA, Head of the Department for providing us with the opportunity and facilities to take up this project.

I am deeply obliged and indebted to the timeless help and guidance provided by Dr. E. SUGANYA, Assisstant Professor, Department of Information Technology and also express my heartfelt thanks for making this project a great success.

I also thank all the faculty of the Department of Information Technology for their kind advice, support and encouragement and last but not the least I thank my parents and my friend for their moral support and valuable help.

REFERENCES

[1] M. Luca, "Reviews, reputation, and revenue: The case of yelp.com," Harvard Business School NOM Unit Working Paper, no. 12-016, 2016.

[2] Y. Wu, E. W. Ngai, P. Wu, and C. Wu, "Fake online reviews: Literature review, synthesis, and directions for future research," Decision Support Systems, vol. 132, p. 113280, 2020.

[3] S. Dixit and B. Ravindran, "Graph-based fake review detection via suspicious pattern discovery," in Proceedings of the 30th ACM International Conference on Information & Knowledge Management, pp. 3216–3220, 2021.

[4] N. Jindal and B. Liu, "Opinion spam and analysis," Proceedings of the 2008 international conference on web search and data mining, pp. 219–230, 2008.

[5] M. Ott, Y. Choi, C. Cardie, and J. T. Hancock, "Finding deceptive opinion spam by any stretch of the imagination," Proceedings of the 49th annual meeting of the association for computational linguistics, pp. 309–319, 2011.

[6] S. Feng, R. Banerjee, and Y. Choi, "Distributional footprints of deceptive product reviews," in Proceedings of the International AAAI Conference on Web and Social Media, vol. 6, pp. 98–105, 2012.

[7] R. Mohawesh, S. Xu, S. Tran, R. Ollington, M. Springer, Y. Jararweh, and S. Maqsood, "Fake reviews detection: A survey," IEEE Access, vol. 9, pp. 65771–65802, 2021.

[8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," arXiv preprint arXiv:1810.04805, 2018.

[9] J. Wang and L. Chen, "An aspect-based sentiment analysis framework examining semantic consistency reviews," Expert Systems with Applications, 2022. https://www.sciencedirect.com/science/article/abs/pii/S0957417422001985.

[10] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[11] M. Gayathri, Y. S. Teja, and K. A. Sharma, "Fake review detection using machine learning," International Journal of Information Technology, 2023.

[12] Y. Kim, "Convolutional neural networks for sentence classification," in Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 1746–1751, 2014.

[13] A. Mukherjee, V. Venkataraman, B. Liu, and N. Glance, "What yelp fake review filter might be doing?," Proceedings of the International AAAI Conference on Web and Social Media, vol. 7, no. 1, pp. 409–418, 2013.

[14] V. Punde et al., "Fake review detection using sentimental analysis," International Journal of Advanced Computer Science and Applications, 2021.

[15] S. Rayana and L. Akoglu, "Collective opinion spam detection: Bridging review networks and metadata," in Proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining, pp. 985–994, 2015.

[16] R. Singh and D. Patel, "A hybrid approach combining nlp with reviewer behavior analysis," Revue d'Intelligence Artificielle, vol. 37, no. 5, 2023. https://www.iieta.org/journals/ria/paper/10.18280/ria.370507.

[17] D. Zhang, L. Zhou, J. L. Kehoe, and I. Y. Kilic, "Enhancing fake review detection: An approach using bidirectional lstm and attention mechanism," Journal of Management Information Systems, vol. 37, no. 4, pp. 1125–1156, 2020.

[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," Advances in neural information processing systems, vol. 30, 2017.

[19] S. Shehnepoor, M. Salehi, R. Farahbakhsh, and N. Crespi, "Netspam: a network-based spam detection framework for reviews in online social media," in Proceedings of the International Conference on Web Intelligence, pp. 991–998, 2017.

[20] K. Crawford, R. Dobbe, T. Dryer, G. Fried, B. Green, E. Kaziunas, A. Kak, V. Mathur, E. McElroy, A. N. Sánchez, et al., "Ai and the everything in the whole wide world benchmark," arXiv preprint arXiv:2111.15366, 2021.

[21] R. Barbado, O. Araque, and C. A. Iglesias, "A framework for fake review detection in online consumer electronics retailers," in Proceedings of the 2019 IEEE/WIC/ACM International Conference on Web Intelligence, pp. 38–45, 2019.

[22] N. Kumar et al., "An innovative time-series analysis method for detecting fake review patterns," IEEE Transactions on Knowledge and Data Engineering, 2020. https://ieeexplore.ieee.org/document/9123672.

[23] Y. Li, X. Zhao, T. Xu, and Y. Zou, "Bert-based hybrid model for fake review detection," in 2022 IEEE 5th International Conference on Information Systems and Computer Aided Education (ICISCAE), pp. 1231–1235, IEEE, 2022.

[24] N. Kumar, D. Venugopal, L. Qiu, and S. Kumar, "A survey on deep learning-based fake review detection models," Information Systems Frontiers, pp. 1–24, 2022.

[25] N. Jindal and B. Liu, "A groundbreaking approach using linguistic patterns and behavioral features to identify fake reviews," Expert Systems with Applications, vol. 118, 2018.

[26] A. Rodriguez et al., "Graph convolutional networks (gcn) integration with semantic similarity measures," Applied Sciences, vol. 13, no. 4, p. 2244, 2023. https://www.mdpi.com/2076-3417/13/4/2244.

[27] J. Thompson et al., "A system leveraging transformer models to distinguish between ai-generated and human-written reviews," IEEE Transactions on Artificial Intelligence, 2024. https://ieeexplore.ieee.org/document/9716933.